

Week 15 Report – Visual CNN Training and Evaluation

Objective

The objective of Week 15 was to train and evaluate visual convolutional neural network (CNN) models for deepfake detection using the FaceForensics++ C23 dataset. The focus was on measuring dataset-specific performance using standard classification metrics.

Models Implemented

Two frame-level visual CNN baselines were evaluated:

1. MobileNetV2 (Baseline)
 - Lightweight architecture optimised for efficiency.
 - Used as an initial benchmark.
2. Xception (Baseline)
 - Deeper CNN architecture designed to capture fine-grained spatial artefacts.
 - Implemented using the timm library with ImageNet pretraining.

Both models output a single binary logit for real vs fake classification.

Training Setup

- Dataset: FaceForensics++ (C23 compression)
- Input: Uniformly sampled video frames
- Resolution:
 - MobileNetV2: 224×224
 - Xception: 299×299
- Loss Function: Binary Cross-Entropy with Logits
- Optimiser: AdamW (learning rate = $1e-4$)
- Epochs: 5
- Hardware: CPU
- Training paradigm: Frame-level supervision

Evaluation Protocol

- valuation level: Video-level (aggregation of frame predictions).
- Test set size: 712 videos.
- Metrics reported:
 - Accuracy
 - F1-score
 - AUC (Area Under ROC Curve)
 - Confusion Matrix
 - Precision / Recall per class

Results (Xception - FF++ C23)

Metric	Value
Accuracy	0.7458
F1-score (fake class)	0.8454
AUC	0.6052
Videos evaluated	712

Confusion Matrix:

[[36 128]

[53 495]]

The model demonstrates robust performance in detecting fake videos, with high recall and F1-score for the fake class, while performance on real videos is lower due to dataset imbalance.

Discussion

The Xception model outperformed the MobileNetV2 baseline, particularly in terms of AUC, indicating improved discrimination capability. However, the bias towards predicting fake content highlights a known limitation of frame-level CNNs trained on imbalanced datasets. These results are consistent with findings in existing deepfake detection literature.

The observed limitations motivate the use of multimodal (audio-visual) and temporal modelling approaches, which will be explored in subsequent stages of the project.

Conclusion

Week 15 objectives were completed. Visual CNN baselines were trained and evaluated on FaceForensics++ C23, and all required metrics (Accuracy, F1, AUC) were obtained and documented. The results establish a solid visual baseline for comparison with future multimodal and temporal models.